

THE ZERO PARADOX

ZP-J Keystone Addendum

The Diagonal Fixed Point: the Lawvere Face-Split and the Well-Foundedness Boundary

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The keystone of the Zero Paradox is that $\perp$ plays the same self-referential (diagonal) fixed-point role in every framework: the Quine atom $\perp = \{\perp\}$ in set theory, the Kleene quine in computation, the point $v_2(0) = \infty$ in valuation, and the initial object in category theory. Their shared membership in that family (MC-1) is proved per domain for the Quine atom, 2-adic and categorical faces; the Kleene face is carried as a requirement, its witness `computability_face_fixedPoint` a fixed point of a computable map, not tied to $\perp$; the choice of criteria is a design principle; and the claim that the faces are numerically one object is retired as ill-typed (object equality across categories does not typecheck and is not invariant under equivalence). This addendum is a thin, honest record of two machine-checked investigations into the structure of that keystone — both probe-level, both fenced as to exactly what they prove.

The first asks whether the keystone is a literal instance of Lawvere's fixed-point theorem (the recognized home of the diagonal family). The answer is face-dependent and machine-checked. The second gives the keystone's structural home in well-founded coalgebra theory (Taylor; Adámek–Milius–Moss): the snap $\perp \rightarrow \epsilon_0$ as a crossing of the well-foundedness boundary. Neither investigation is claimed beyond what Lean verifies; the deeper category-theoretic result is cited, not re-proved.

Section I: The Keystone and its Recognized Home

The unification of self-referential fixed points across fields is not new with the Zero Paradox. Lawvere (Diagonal Arguments and Cartesian Closed Categories, 1969) showed that Cantor's diagonal, Russell's paradox, Gödel's incompleteness lemma and Tarski's undefinability theorem are one move; Yanofsky (A Universal Approach to Self-Referential Paradoxes, Incompleteness and Fixed Points, Bull. Symbolic Logic 9(3), 2003) restated this in plain set-and-function terms across logic and computation, adding Turing's halting argument and the recursion theorem. Those faces are prior art, cited and not claimed.

What the Zero Paradox adds (and what it does not)

Adds: candidate faces outside the classical scheme — the 2-adic valuation $v_2(0) = \infty, \epsilon_0$, the wheel of fractions — each with a machine-checked axiom footprint; and the location claim, that the fixed point sits at the floor $\perp$ (the Gödel inversion), a framing.

What the Zero Paradox adds (and what it does not)

Does not add: the unification itself (Lawvere/Yanofsky), nor any identification of the four faces as one object — that identity is retired as ill-typed (object equality across categories does not typecheck and is not invariant under equivalence). What MC-1 carries is family membership, proved per domain for the Quine atom, 2-adic and categorical faces (the Kleene face carried as a requirement).

Section II: The Lawvere Face-Split (machine-checked)

Lawvere's theorem (Mathlib's `Function.exists_fixed_point_of_surjective`) says a point-surjection $\alpha \rightarrow (\alpha \rightarrow \beta)$ — the diagonal hypothesis — forces every endofunction of β to have a fixed point. The keystone's fixed points are posited (the self-application has $\perp$ as its fixed point); Lawvere's are derived from a surjection. So the question is whether each face supplies that surjection. The verdict is face-dependent.

In Set: no face is a Lawvere instance (Lawvere.lean)

For any nontrivial type, a Lawvere witness cannot exist: it would force every endofunction to have a fixed point, impossible once there are two distinct elements (Cantor).

`nontrivial_lattice_no_witness` — the lattice / set-theoretic face: no witness on a nontrivial $\perp$ -pointed lattice.

`q2_no_witness` — the 2-adic face: no witness on $\mathbb{Q}_2$.

So in Set the keystone's $\perp$ is a posited fixed point of one specific self-map, not a Lawvere-derived one. The "diagonal fixed point" name is an analogy at this face.

`fixedPoint_of_witness`, `no_witness_of_fixedPointFree`: fully axiom-free.

In computability: a genuine instance (Lawvere.lean)

`computability_face_fixedPoint` — every computable self-map on codes has a fixed point up to eval: two codes computing the same function. This wraps Mathlib's `Nat.Partrec.Code.fixed_point` (Rogers / Kleene's recursion theorem).

The qualifier is load-bearing. A literal fixed point is not on offer — $c \mapsto \text{pair}(c, c)$ is total, computable, and returns its own input for no c . The escape from the Cantor obstruction is a restriction on which MAPS exist, not on which type eval lands in: no computable self-map on codes is eval-fixed-point-free, so the diagonal that would refute the witness has no computable representative and the obstruction cannot fire. This is ZP-K's face (the Kleene quine).

Footprint: `[propext, Classical.choice, Quot.sound]`, inherited from Mathlib.

The verdict, plainly: the test is category-relative. In Set no face is a Lawvere instance; in the effective (computability) category the recursion theorem is a genuine one. The keystone therefore unifies a shape (the diagonal), not a single mechanism. What MC-1 carries is family membership, proved per domain for the Quine atom, 2-adic and categorical faces (the Kleene face carried as a requirement), with the choice of criteria the design commitment.

Section III: The Well-Foundedness Boundary

The keystone's structural home is well-founded coalgebra theory. A relation is encoded as a powerset coalgebra; well-founded means no infinite descending chain — no cycle. The Quine atom $\perp = \{\perp\}$ is exactly the minimal non-well-founded coalgebra: the self-loop $\perp \in \perp$, the back edge. Taylor (Well-founded coalgebras and recursion) proves well-founded $\Rightarrow$ recursive as his Recursion Theorem (Thm 36, p. 15), and states the converse in a topos — there well-foundedness is necessary for recursion (Prop 111, p. 6). Adámek–Milius–Moss (On Well-Founded and Recursive Coalgebras, 2020, arXiv:1910.09401v2) prove the forward direction as the General Recursion Theorem (Thm 7.2, p. 27). Their converse (§ 8) always asks the endofunctor to preserve inverse images, and then takes one of several routes: the category may have universally smooth monomorphisms with the functor carrying a pre-fixed point (Thm 8.1), or the category may have a subobject classifier (Thm 8.6, which is Taylor's); a third covers functors on vector spaces, which have neither (Thm 8.12). Under Thm 8.1's assumptions the characterizations coincide — well-founded $\Leftrightarrow$ recursive $\Leftrightarrow$ a morphism to the initial algebra (Cor 8.2). Reading the necessity direction as one cannot recurse through $\perp$ is this framework's gloss, not Taylor's: Prop 111 names no bottom element.

In that language the binary snap $\perp \rightarrow \epsilon_0$ is a crossing of the well-foundedness boundary: from the non-well-founded floor ($\perp$, the self-loop, where recursion cannot reach) to the well-founded ascent (the ϵ_0 ordinal tower, recursively generated). The Zero Paradox formalizes this at two levels.

Relation level (Boundary.lean)

floor_not_wellFounded — the self-application floor is non-well-founded ($\perp$ self-loops under selfApp). Fully axiom-free.

ascent_wellFounded — the ordinal ascent is well-founded (Ordinal.lt_wf).

snap_crossing — on a single carrier, the floor is the sole non-accessible point; every post-snap state is accessible. The snap exits the unique non-well-founded point into the well-founded ascent.

Categorical bridge (BoundaryBridge.lean)

snap_boundary_two_registers — the crossing witnessed in two registers at once: the relation/carrier level above, and ZP-P's categorical μ/ν fork (categorical_fork_strict: the initial algebra empty, the final coalgebra inhabited — the self-referential element lives in ν , not μ).

Honest scope — what is proved, and what is committed

Proved: the relation-level boundary and the QPF μ/ν bridge above (floor_not_wellFounded is axiom-free; the rest carry Classical.choice from Mathlib's ordinal and QPF machinery).

No new commitment: that the snap is this crossing is a faithful model whose content is the two proven endpoints, plus the reading of those endpoints as ZP's own $\perp$ and ϵ_0 respectively — a role assignment the framework already carries (MC-1 family membership; the ϵ_0 type bridge open under OQ-E2), not a fresh one. The two endpoints stay distinct: $\epsilon_0 \neq \perp$ (epsilon0_ne_bot). The floor endpoint is tied to the real $\perp$ of ZP (floor_not_wellFounded, axiom-free); the single-carrier Phase is the illustrative toy model, where non-well-foundedness localizes at the floor by construction.

Section IV: What Is Not Formalized (and an Open Invitation)

The full Taylor coalgebraic statement — $\perp$ as a non-well-founded coalgebra in the broken-pullback sense, with the General Recursion Theorem and its converse — is deliberately not formalized here. Searched as of August 2026, the pinned Mathlib carries neither Pataaraia's fixed-point theorem nor a recursion theorem for well-founded coalgebras (classically, the least fixed point of Taylor's Thm 118 and the Pataaraia induction of his Cor. 119 follow from Bourbaki–Witt, as the contribution point below records); the next-time operator on subobject lattices is no longer missing, having since been built in this project (ZeroParadox/Category/NextTimeCategorical.lean), though it is not upstreamed. The depth result is therefore cited (Taylor; Adámek–Milius–Moss), not re-proved. What is given here is the relation-level boundary and the QPF bridge: a best effort that names its own boundary.

Open contribution point

Classically, the two things Taylor needs from Pataaraia — the least fixed point of a monotone self-map of a directed-complete order with a least element (Taylor Thm 118), and Pataaraia induction (Taylor Cor. 119) — follow from the Bourbaki–Witt theorem, which the pinned Mathlib carries (Mathlib/Order/BourbakiWitt.lean). That route is standard: a monotone map on a chain-complete order has a least fixed point (Zermelo's theorem, not constructive; Adámek–Milius–Moss, CALCO 2021, Thm 2.1), and a directed-complete order with a least element is chain-complete (Bauer–Lumsdaine, arXiv:1201.0340, p. 3). This project's Lean instance of it is `dcpo_exists_least_fixedPoint` and `pataaraia_induction` (ZeroParadox/Order/PataaraiaFromBourbakiWitt.lean), with `Classical.choice` in their footprint. Dropping choice is a separate matter. Classical logic without choice suffices (Dubut and Yamada's Isabelle/HOL library, checked with the axiom of choice excluded, generalizes Pataaraia's theorem; LMCS 2022), but Lean derives excluded middle from `Classical.choice`, so a choice-free Lean version would follow an intuitionistic proof such as Pataaraia's (Taylor Prop 117 with Thm 118). That proof is formalized in Agda (TypeTopology, Various.Pataaraia and Various.Pataaraia-Taylor, 2024); a Lean port was not located in the pinned Mathlib or Lean core as of September 2026. What upgrades this best-effort bridge to the full coalgebraic statement is the General Recursion Theorem, with well-founded coalgebras; porting the intuitionistic proof would additionally take `Classical.choice` off the fixed-point step. Each would be a reusable Lean contribution independent of the Zero Paradox. Upstreaming the next-time operator, already built here, is a third. Contributions are welcome, to this project and to Mathlib as a whole; the precise missing pieces are named above.

Endnote: This document is an addendum to ZP-J Self-Reference and reads after it. ZP-J established the porthole ($v_2(\perp) = \infty$, $\perp = \{\perp\}$); this addendum records two machine-checked probes into the keystone's structure — the Lawvere face-split and the well-foundedness boundary — each fenced as to exactly what it proves. Lean sources: ZeroParadox/Category/Lawvere.lean, ZeroParadox/Multihomed/Boundary.lean, ZeroParadox/Multihomed/BoundaryBridge.lean, ZeroParadox/Order/PataaraiaFromBourbakiWitt.lean, all sorry-free in Lean 4 as of September 2026.